

GC-7: AC + GC as Dual Tools — The Complete Proof Strategy

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Abstract

Arithmetic Complexity (AC) and Geometric Constraint (GC) are dual proof tools. AC classifies the computational depth of a derivation — how many sequential counting steps it requires. GC identifies the structural content — the independent constraints that force a unique answer. Neither is sufficient alone. AC without GC tells you a problem is tractable but not what the answer is. GC without AC tells you the answer but not whether it is derivable at bounded depth. Together they constitute a complete proof strategy: bounded-depth derivation of unique structures. This note formalizes the pairing, tabulates how it operates across the Millennium proofs and beyond, and identifies the conditions under which it applies. The synthesis provides the theoretical foundation for GC-9 (the methodology paper).

1. The Two Tools

1.1 AC: The Depth Bound

Arithmetic Complexity classifies every mathematical statement by its AC(0) depth — the length of the longest sequential chain of genuine counting operations (T96, T316, T421). The key definitions:

- Depth 0: Definitions, identities, substitutions, invocations of proved results, comparisons, table lookups. These are free. They are the wiring of a proof, not its computational content.
- Depth 1: One genuine counting step — a summation over a new index set, a spectral integration along one direction of the maximal flat in D_{IV}^5 .
- Depth 2: Two sequential counting steps where the second depends on the output of the first. Each traverses an independent direction in the rank-2 spectral parameter space.
- Depth 3+: Does not exist for D_{IV}^5 . The rank is 2; there are exactly two orthogonal spectral directions. A third sequential count would require a third independent direction that the geometry does not provide (T316, Section 12 three-lemma closure).

The Depth Ceiling Theorem (T316/T421): every mathematical theorem has AC(0) depth at most $\text{rank}(D_{IV}^5) = 2$. Empirical confirmation: 1135+ theorems surveyed, zero depth-3 results. The Classification of Finite Simple Groups (10,000 pages) is depth 2. Godel's First Incompleteness is depth 1. The depth ceiling is not a limitation — it is the structure of computation on this geometry.

Under the stronger Casey strict criterion (T421), depth collapses further: bounded enumeration is depth 0, eigenvalue identification is depth 0, Fubini collapse makes double integrals depth 0. The effective ceiling becomes depth 1 for most results.

1.2 GC: The Uniqueness Engine

The Geometric Constraint method (GC-5) provides the structural content of a proof. Its three-move version:

1. Constraint: Find independent bounds that pin the structure. A lower bound (what the problem MUST satisfy) and an upper bound (what mathematics CAN reach) meet with zero room. The answer is forced.
2. Certificate: Verify computationally that the constraint is tight — run all candidates in the relevant classification, confirm only one survives.
3. Boundary: State explicitly what you did NOT prove.

The five-step implementation (GC-5, Section 1b) expands this to: Constructive Constraint, Exclusion Lemmas, Cross-Type Cascade, Over-Determination, Honest Scope. Steps 1-3 prove the theorem; step 4 measures evidence quality; step 5 distinguishes proof from claim.

GC identifies the *what*: these specific constraints, this unique solution, this scope of applicability.

1.3 Why Neither Is Sufficient Alone

AC without GC: You know the problem can be solved at depth d . You know a proof exists that requires at most d sequential counting steps. But you do not know what the proof says. The depth classification is a complexity measure, not a derivation. Knowing that the Hodge conjecture is depth 1 does not tell you what the theta lift is, which domain survives, or why D_{IV}^5 is unique. AC provides the computational budget. It does not spend it.

GC without AC: You have the answer — the unique structure forced by n independent constraints. But you do not know whether the derivation is tractable. A conjecture could be forced by constraints and yet require unbounded depth to verify. GC provides the geometric content. It does not guarantee that the content is derivable in bounded operations.

AC + GC: The answer is both derivable AND unique. The derivation has bounded depth (AC), and the result is the only one compatible with the constraints (GC). This is what “proved” means in the BST framework: a bounded-depth derivation of a forced structure.

2. The Pairing in Practice

For each Millennium proof, the pairing operates as follows. AC assigns the depth — how many sequential counting steps the proof requires. GC identifies the constraints — what pins the answer. The proof is the combination: bounded-depth derivation of a forced structure.

2.1 The AC Depth Table for GC Proofs

Proof	AC Depth	Count 1 (Constraint)	Count 2 (Resolution)	GC Constraints	Over-Det Ratio
Four-Color	0	Structural induction (D0)	—	Forced fan lemma	~5:1
Hodge	1	Theta lift (one spectral integration)	—	5 constraints, squeeze $n_C=5$	6.6:1
YM	1	Spectral construction (Bergman gap)	—	5 constraints, gauge-matter split	9.4:1
RH	2	Spectral gap + multiplicity ($m_s=3$)	Weyl enumeration (wall projection)	4 spectral filters	5:1
BSD	2	Chern hole (topological obstruction)	Transfer chain (1/rank universality)	Modularity + Chern + 1/rank	6.6:1
NS	2	Enstrophy bound (solid angle)	Blow-up ODE ($P \geq c \cdot \Omega^{3/2}$)	$N_{\text{eff}} \leq 5$, $P > 0$	3:1
$P \neq NP$	2	Parity erasure (block independence)	Godel trichotomy (BSW on width)	3 independent routes	3 routes
Poincare	2	Entropy monotonicity (W-entropy)	Finite extinction (width $\rightarrow 0$)	Thurston classification	—
FLT	2	Ribet level-lowering ($N \rightarrow N/q$)	Dimension count ($S_2(\Gamma_0(2))=0$)	Modularity lifting	—

2.2 Reading the Table

Depth 0 (Four-Color): No genuine counting step is needed. The proof is entirely structural — definitions and induction. GC provides the fan lemma; AC confirms the derivation is depth 0. The simplest case: forced structure, zero computation.

Depth 1 (Hodge, YM): One genuine counting step. For Hodge: the theta lift is one spectral integration — enumerate representations, verify saturation. For YM: the Bergman spectral construction is one spectral evaluation. In both cases, GC provides the algebraic squeeze ($n_C \geq 5$ AND $n_C \leq 5$), and AC confirms the squeeze requires exactly one count. The constraint is found and verified in a single pass.

Depth 2 (RH, BSD, NS, $P \neq NP$, Poincare, FLT): Two sequential counting steps, matching the pattern identified in the Depth Ceiling paper (Section 3): Count 1 identifies the obstruction, Count 2 resolves it. GC provides the specific constraints for each problem; AC confirms the obstruction-resolution pair requires exactly two sequential counts and no more.

2.3 The Pattern

In every case: - AC depth tells you the architecture: how many sequential stages the proof needs. - GC constraints tell you the content: what specific mathematical objects populate each stage. - The proof is the pairing: the architecture instantiated with content.

A depth-1 proof has one GC constraint that can be verified in a single spectral pass. A depth-2 proof has a paired obstruction: the first count finds it, the second resolves it — and the specific obstruction is identified by GC's constraint/certificate/boundary method.

3. Why the Pairing Works

3.1 Complementary Roles

AC and GC partition the proof task cleanly:

Aspect	AC Provides	GC Provides
Complexity	Depth bound ($d \leq 2$)	Not addressed
Uniqueness	Not addressed	Constraint count, over-determination ratio
Content	What operations are performed	What objects are operated on
Verification	Bounded-depth decidability	Cross-type cascade (computational certificate)
Scope	Applicability class (spectral, combinatorial)	Honest boundary (what is NOT proved)
Quality	Depth as parsimony measure	Over-determination ratio as robustness measure

There is no overlap. Each addresses exactly what the other leaves open.

3.2 The (C,D) Framework as Bridge

The (C,D) framework (T422) provides the formal bridge between AC and GC. Every theorem receives two labels:

- C (conflation): How many parallel subproblems are conflated into the statement. This is width — it can be arbitrarily large without affecting depth.
- D (depth): How many sequential counting steps are required. This is the AC depth.

GC's over-determination ratio maps to C: a ratio of 6.6:1 means 33 constraints pin 5 parameters — the conflation count is the number of constraint sources. AC's depth maps to D: the sequential chain length.

The (C,D) label is the complete signature of a theorem's proof complexity. C tells you how wide the GC analysis is. D tells you how deep the AC chain is. A theorem with (C=33, D=1) — like the Hodge proof — has enormous constraint width but minimal computational depth. The difficulty was in *finding* the 33 constraints (a creative act, depth 0), not in *verifying* them (one counting step, depth 1).

3.3 Formal Statement

Theorem (AC+GC Completeness, informal). Let C be a GC-amenable conjecture (GC-5, Section 4). Let d be the AC(0) depth of the proof of C. Then:

1. The proof of C consists of d sequential counting steps, each operating on structures identified by GC constraints.
2. The GC constraints provide the input to each counting step: what to count, what bounds to verify, what exclusions to check.
3. The AC depth provides the output guarantee: the computation terminates, the answer is decidable, the derivation is bounded.
4. The combination is a proof: a bounded-depth derivation (AC) of a unique structure (GC).

The theorem is “informal” because formalizing it requires a meta-mathematical claim (T147, BST-AC Structural Isomorphism) that all mathematics reduces to spectral analysis on D_{IV}^5 . This is the unconditional scope question identified in the Depth Ceiling paper (Section 12.4). The conditional version — for theorems expressible as spectral queries on D_{IV}^5 — is proved.

4. The Depth Anatomy

4.1 What Makes a Problem Depth 0

A depth-0 proof requires no genuine counting. The answer follows from definitions, identities, and structural induction. In GC terms: the constraints are purely topological or combinatorial — they can be verified by inspection, not enumeration.

The Four-Color Theorem is depth 0 because the forced fan lemma is structural: no spectral integration, no enumeration of candidates requiring summation. The fan closes by planarity + cubicity + bridge-freedom. These are properties, not counts.

AC says: zero computation required beyond definitions. GC says: the structure is forced by geometric necessity. Together: a structural proof — the simplest and most robust kind.

4.2 What Makes a Problem Depth 1

A depth-1 proof requires one genuine count — one spectral integration, one enumeration of an index set. In GC terms: the constraints have a unique solution, but verifying uniqueness requires one pass through the candidate space.

Hodge and YM are depth 1 because the algebraic squeeze ($n_C \geq 5$ AND $n_C \leq 5$) requires verifying that the lower bound actually holds (one count: check unitarity/confinement across representations) and that the upper bound actually holds (one count: check Selberg degree/scattering factorization). But these two verifications are independent — they operate on different constraint sources and can run in parallel. Parallel counts do not add depth (T316, Section 4.3). So the total depth is $\max(1,1) = 1$.

AC says: one counting step, parallelizable. GC says: the squeeze has exactly one survivor. Together: a single-pass verification of a forced answer.

4.3 What Makes a Problem Depth 2

A depth-2 proof requires two sequential counts where the second depends on the first. In GC terms: the constraints have a paired structure — an obstruction that must be found (Count 1) before it can be resolved (Count 2).

The RH is depth 2 because multiplicity counting ($m_s = 3$, the ratio 1:3:5) must complete before Weyl enumeration can use that multiplicity to isolate the dominant term. The BSD is depth 2 because the Chern hole must be established before the transfer chain can use it to link L-function order to algebraic rank. In each case, Count 2's input is Count 1's output. The dependency is genuine and irreducible.

AC says: two sequential steps, neither parallelizable with the other. GC says: paired obstruction — find it, then resolve it. Together: the maximum complexity permitted by the geometry.

5. What the Pairing Teaches

5.1 Difficulty Is Width, Not Depth

The CFSG is 10,000 pages. Its AC depth is 2. Its GC width (conflation count) is enormous — hundreds of case families, thousands of individual groups. The difficulty of CFSG is not that its proof chain is deep; it is that its constraint verification is wide. AC+GC makes this distinction precise: $D=2$, $C=O(10,000)$.

The same applies to every hard problem. Wiles' proof of FLT is depth 2, width ~500 pages. Perelman's geometrization is depth 2, width ~300 pages. The creative work — the work that takes decades and earns Fields Medals — is in the depth-0 layer: constructing the right definitions that make the depth-1 or depth-2 counting steps tractable. Finding the Frey curve. Defining Ricci flow with surgery. Constructing the theta lift.

AC+GC reveals: mathematical difficulty is creative (finding definitions, depth 0) and exhaustive (verifying constraints, width), not computational (sequential depth).

5.2 Over-Determination as Quality Measure

GC's over-determination ratio and AC's depth bound are independent quality measures:

- High ratio, low depth (Hodge: 6.6:1, $D=1$; YM: 9.4:1, $D=1$): The proof is robust (many constraints) and simple (one counting step). This is the gold standard.
- High ratio, higher depth (RH: 5:1, $D=2$; BSD: 6.6:1, $D=2$): The proof is robust but requires a paired obstruction. Still strong — the depth reflects genuine sequential structure, not weakness.
- Lower ratio, higher depth (NS: 3:1, $D=2$): The proof is adequate but less over-determined. This is where the physics dependence shows — NS has fewer purely mathematical constraints.

The ideal proof has high over-determination and low depth: many independent reasons why the answer is right, computed in one step.

5.3 The Scope Contract

AC provides a scope guarantee: every result within the D_{IV}^5 spectral framework has depth at most 2. GC provides a scope limitation: each proof applies only to its stated arena (GC-5, Step 5). The combination is a scope contract:

- AC guarantees: if a proof exists, it is bounded-depth.
- GC specifies: exactly where the proof applies and where it does not.

No proof claims more than its scope. No proof requires more than two counting steps. The contract is symmetric: honest depth, honest boundary.

6. Conditions for Applicability

The AC+GC pairing works when both tools apply simultaneously. This requires:

1. AC applicability: The problem reduces to operations on structures derivable from D_{IV}^5 — spectral integration, counting, enumeration. This is the scope of T147 (BST-AC Structural Isomorphism).
2. GC amenability (GC-5, Section 4): The conjecture concerns a property of structures on a classifiable arena; the arena admits a finite classification; independent constraints over-determine the parameters.
3. Non-degeneracy: The GC constraints actually engage the AC counting — the verification is not trivially depth 0. (If it is, the proof is purely structural and AC contributes only the classification “depth 0.”)

Problems that satisfy (1) but not (2) — for example, probabilistic statements, pure existence theorems — can be depth-classified by AC but not structurally pinned by GC. Problems that satisfy (2) but are outside the current scope of (1) await the extension of T147.

The seven Millennium proofs, plus FLT and Poincare, all satisfy (1)-(3). This is not coincidental. The Millennium problems were selected because they are fundamental — and fundamental problems tend to be both spectrally expressible and structurally forced.

7. Summary

Property	AC Alone	GC Alone	AC + GC
Tells you	Depth of derivation	Uniqueness of answer	Both
Guarantees	Bounded computation	Forced structure	Complete proof
Measure	Depth d in $\{0,1,2\}$	Over-determination ratio R	(C,D) label
Limitation	No content (what)	No complexity (how fast)	Scope of T147
Signature	“Computable”	“Unique”	“Proved”

The central claim: AC and GC are dual tools whose pairing constitutes a complete proof strategy for GC-amenable conjectures within the D_{IV}^5 spectral framework. AC provides the computational guarantee (bounded depth). GC provides the structural guarantee (unique answer). Together they deliver proofs that are both tractable and forced — the hallmark of deep mathematics done simply.

References

- GC-5: notes/BST_GC5_Five_Step_Methodology.md — Three-move and five-step GC methodology
- T96 (Depth Reduction Lemma): Composition with definitions is free
- T316 (Depth Ceiling): AC depth $\leq \text{rank}(D_{IV}^5) = 2$
- T421 (Depth-1 Ceiling): Under Casey strict, depth ≤ 1
- T422 ((C,D) Framework): Conflation and depth as independent measures
- T147 (BST-AC Structural Isomorphism): Force+boundary = counting+boundary
- Depth Ceiling paper: notes/BST_AC_DepthCeiling.md
- AC Theorem Registry: notes/BST_AC_Theorem_Registry.md

This note provides the theoretical foundation for GC-9 (the methodology paper).